

CSMA/CD & CSMA/CA Protocol:-

Carrier sense multiple access with collision Detection.

→ CSMA/CD detects collision & attempts to recover from them by stopping transmission & retrying later.

• Carrier Sense:-

Before transmitting data, a device using CSMA/CD listens to network to check if channel is idle. If it senses that the channel is busy, it waits for it to become idle.

• Multiple access:-

Multiple devices share same channel

• Collision Detection:-

If two devices happen to transmit data simultaneously, causing a collision, CSMA/CD detects this collision. Upon collision detection, both transmitting devices stop send^g data & wait for a random amt of time before attempting to retransmit.

CSMA/CA (Collision Avoidance)

- CSMA/CA aims to avoid collisions by coordinating transmissions through RTS/CTS packets.
- Carrier sense:-
Similar to CSMA/CD, devices listen to network to check if channel is idle before transmitting.
- Multiple Access:-
Multiple devices share same channel.
- Collision Avoidance:-
Unlike CSMA/CD, where collisions are detected after they occur, CSMA/CA attempts to avoid collision together. Before transmitting a device sends a small request to send (RTS) packet to recipient.
If recipient acknowledges this packet with a clear to send (CTS) packet, the sender can proceed with transmitting data.
This helps in avoiding collision.

ALOHA Protocol

- Aloha, whether pure or slotted, is a simple protocol where stations transmit whenever they have data, with mechanisms for dealing with collisions through retransmissions.
- It is multiple access protocol used in shared networks.
- It has 2 types:- Pure & slotted.
- In pure, any station can transmit data whenever it has data to send.
- When a station wants to send data, it waits for next time slot (frame) & send its data. After send^g, it waits for an acknowledgement from the receiver.
- If acknowledgement is not received within a certain time frame, the station assumes a collision has occurred & retransmits data after random backoff time.
- In slotted, the time is divided into fixed-size slots, and stations are synchronized to these slots. This reduces chances of collision as transmission occurs only at the beginning of each time slot.

Architecture of IEEE 802.11

→ commonly known as Wi-Fi, specifies the protocols and technologies for wireless local area networking (WLAN) communicatⁿ.

Architecture:-

1) Physical layer:-

- defines physical char. of wireless transmission medium, includes frequencies, modulation schemes, & transmission rates.
- diff. variations of standard (eg 802.11a, 802.11b, 802.11g, etc). operate in diff. freq. bands

2) Medium Access Control (MAC) layer:-

- Responsible for managing access to shared wireless medium
- Includes mechanisms for addressing, packet fragmentation, encryption, error detection
- supports multiple access methods

3) Service Sets:-

- define diff. type of wireless networks within IEEE 802.11 std.
- Basic service set (BSS):-
consist of wireless stations communicating directly with each other within a single access point (AP)
- Extended service set (ESS):-
consists of multiple BSSs interconnected by distributⁿ system

4) Access points

- serves as central hubs for wireless clients to connect to WLAN.
- Manage network access, authentication, encryption & traffic distribⁿ within BSS
- Bridges data betⁿ wireless & wired networks.

* Compare diff Wireless Tech:-

IEEE 802.11 (Wifi)	Bluetooth	Cellular Networks
• It has short to medium range (upto 100 m)	• short range (upto 10 m)	• wide area coverage (upto several kms)
• Data rate is high (upto several Gbps)	• Data rate is moderate (upto 2 mbps)	• high (hundred mbps)
• power consumption moderate to high	• low	• moderate to high

Flow & Error Control in DLC.

* Flow control:-

Data Link control

used to manage flow of data betⁿ sender and receiver to ensure that sender does not overwhelm receiver with data it cannot process.

Two common techniques:-

1) Stop and - wait:-

After send^g a data frame, sender waits for an acknowledgement (ACK) from receiver before send^g next frame.

2) Sliding window:-

Allows sender to transmit multiple frames before waiting for acknowledgements.

* Error control:-

used to ensure integrity of data transmission by detecting & correcting errors that may occur during transmission.

Techniques:-

1) Checksum:-

Sender calculates a checksum based on data being transmitted & include it in frame. receiver recalculates checksum upon receiv^g frame & compares it to checksum sent by sender.

2) Acknowledgement & Retransmission:-

After sending a data frame, sender waits for an acknowledgement (ACK) from the receiver. If an ACK is not received within a specified timeout period, sender assumes frame was lost or damaged & retransmit it.

Issues in Data link layer :-

Frame Synchronization

1) Error detection & ~~error correction~~:-

Ensuring receiver can correctly identify the start and end of each frame is crucial for proper data transmission. If frame syn is lost due to noise, receiver may misinterpret data.

2) Error correct & Detect:-

Detecting & correcting errors that occur during transmission is essential for maintaining data integrity. However, error detection & correction mechanisms add overhead to transmission process & may impact performance.

3) Efficiency:-

Ensuring efficient use of available bandwidth & minimizing overhead is a challenge in Data link layer.

4) Duplexing:-

Managing communication in half-duplex or full duplex modes presents challenges in Data link layer.

Cyclic Redundancy Check (CRC)

→ method used in error detection to ensure integrity of data transmitted over a network or stored in storage device.

It works by generating a checksum based on data being transmitted, which is then appended to the data.

= Receiver performs same calcⁿ & compares received checksum with calc^d one.

= If they match, it indicates that data is likely to be error-free.

Here's an eg. of CRC calcⁿ using polynomial divⁿ.

We want to calc a CRC checksum for data '110101' using divisor polynomial '1011'.

if Data = 110101

& divisor poly = 1011

1011	110101 (data)	110101
	- 1011 (divisor)	
	1001	1001
	1001	110
	- 1011	
	110	101
	- 1011	
	101 (remainder)	

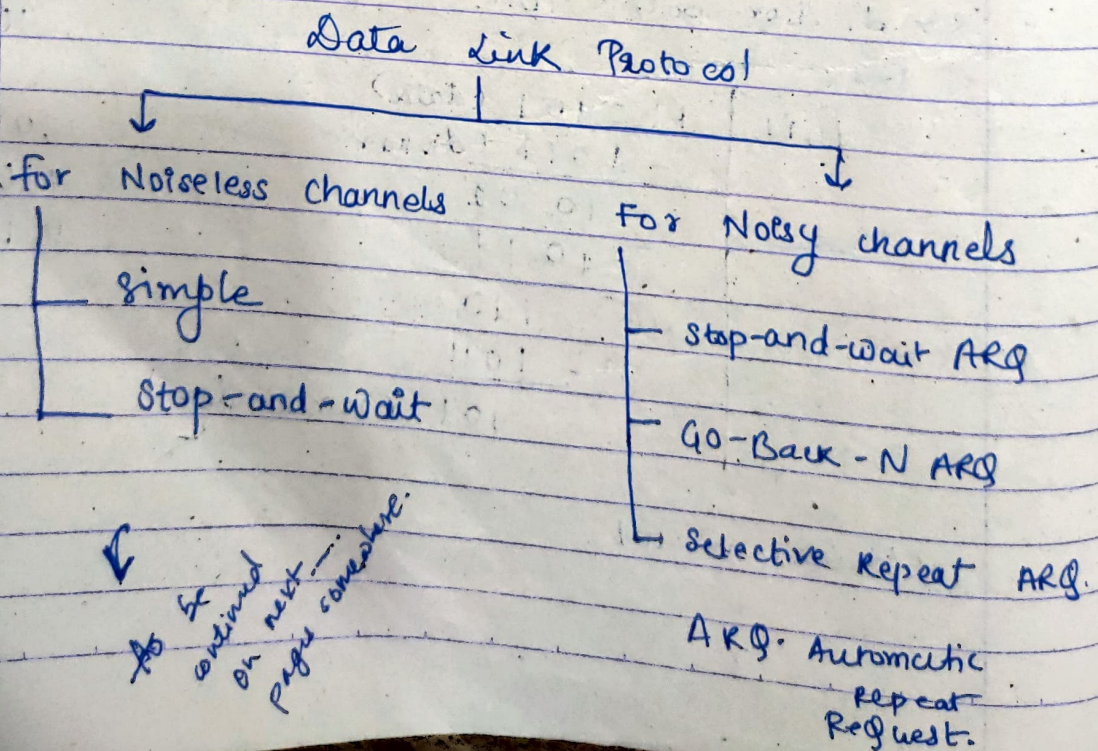
∴ Checksum = 101

Now let's say receiver receives data
 "110101101" with checksum 101.

$$\begin{array}{r}
 1011 \overline{) 110101101} \\
 \underline{- 1011} \\
 1001 \\
 \underline{- 1011} \\
 110 \\
 \underline{- 1011} \\
 101 \text{ (rem)}
 \end{array}$$

Since rem is zero, it seems data is likely to be error-free.
 If rem were nonzero, it would suggest errors may be present.

Q Explain elementary data link protocol with suitable diagrams.



Token Bus

→ • Topology used is Linear Bus Topo

• Token passing direction is bidirectional

• Device access network based on priority

• may suffer from collisions

• More difficult to scale

• coaxial cable is used

• does not keep routing details

• less exp than Ring

Token Ring

• Ring Topology

• unidirectional, (clockwise)

• Device access network based on their posⁿ in ring

• generally offer higher performance

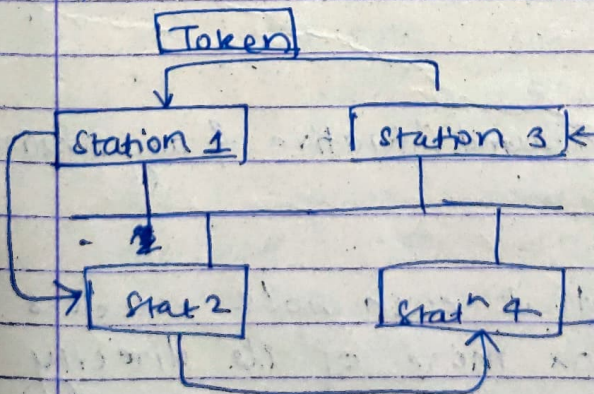
• Easier to scale

• twisted pair & fibre optic

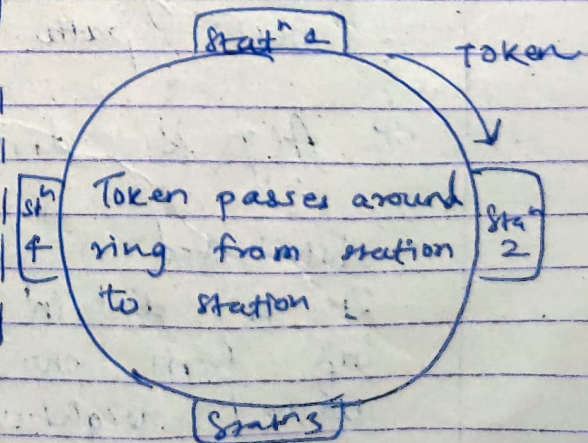
• keeps info of routing

• Expensive

Token Bus:-



Token Ring:-



Classful Addressing

- addresses are divided into classes: A, B, C, D, E.

- does not support subnetting

- Inefficient use of IP address space

- class-based allocation

- eg - class A address range is 1.0.0.0 to 126.0.0.0.

Classless Addressing

- Addresses are not divided into classes.

- support subnetting

- efficient use of IP address space

- CIDR based allocation (Classless Inter-Domain Routing)

- eg - 192.168.10/24 is classless address with network prefix of 24 bits.

Distance Vector Routing algorithm.

→ 1) uses distance as metric to determine best path between two nodes.

2) used in simple network topologies where no. of hops between two nodes is primarily metric used to determine best path.

3) Alg is distributed, iterative & asynchronous.

• distributed:

It is distributed that each node receives info from one or more of its directly attached neighbors & performs calculation.

It then distributes result back to its neighbors.

• Iterative:

process continues until no more info is available to be exchanged betⁿ neighbor.

• Asynchronous

does not require that all of its nodes operate in lock step with each other.

Limitation:

1) Slow convergence :- (S.C)

- have S.C because routers only exchange routing info with their directly connected neighbors.

- It may take several iterations for all routers in network to converge on best route.

2) Routing loops :-

- If there are inconsistencies or delays in updating routing tables, routers may choose suboptimal paths or create loops where packets continuously circulate betⁿ routers without reach^g their destinatⁿ. - It waste network resources but also degrade performance.

* Broadcast, Multiple Cast & Hierarchical routing:-

① Hierarchical:-

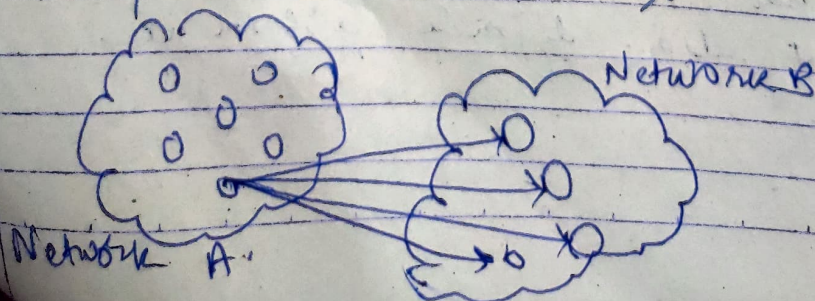
- ⇒ - In... , routers are divided into regions. Each router has complete details about how to route packets to destinations within its own region.
- efficient as routing decisions are made based on hierarchical addressing, reducing scope of routing updates.
 - highly scalable
 - eg:-

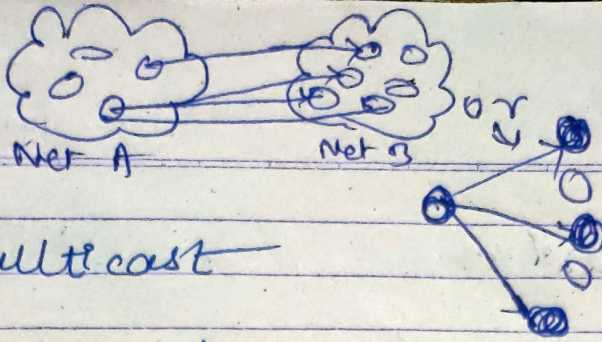
Border Gateway Protocol (BGP) uses H.R. to efficiently route

② Broadcast:-

- data
- Sends packets to all devices in the network.
 - uses a broadcast address to reach all devices in the network.
 - may suffer scalability issues in large networks due to excessive broadcast traffic.

Eg:- Ethernet networks often use broadcast routing for address resolution protocol like (ARP).





③ Multicast

- sends data packets to specific grp of devices.
- uses multicast group address to reach specific group of devices.
- efficient, more scalable than broadcast.
- e.g. - Internet-based streaming services often use multicast routing to deliver content to multiple subscribers.

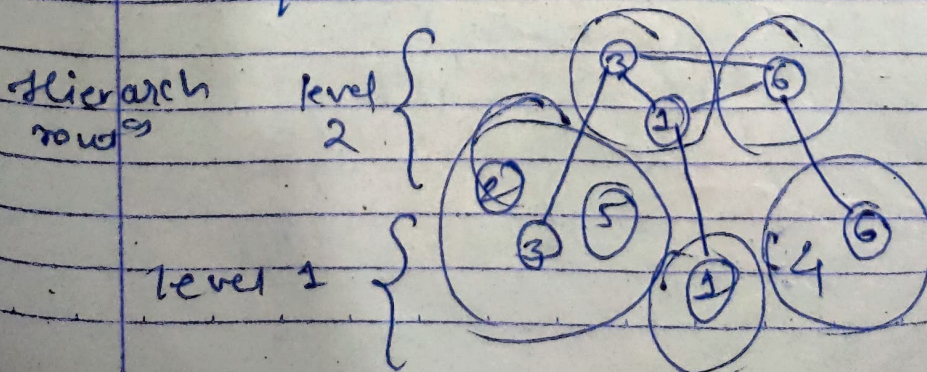
• Short Note:-

① choke Packets:

- special control packets sent by routers to manage network congestion
- instruct sender to reduce its transmission rate temporarily
- help prevent network congestion

② Load shedding:-

- used to manage network congestion by selectively dropping packets during periods of high traffic.
- ensure essential services receive adequate bandwidth.



Jitter Control:-

- refers to variation in packet arrival times in a network.
- aim to minimize packet delay variation to ensure smooth & predictable data transmission.
- Techniques used to mitigate jitter are buffering, packet scheduling, etc.

Traffic Shaping:-

- technique used to regulate flow of network traffic to ensure optimal performance.

Smoothens traffic patterns by controlling rate at which packets are transmitted.

Checksum Technique

100 1100 | 1110 00 10 | 00100100 | 100 00100
 1 2 3 4 ✓

k = 4
 m = 8 bits

Sender :-

1 → 10011001

2 → 11100010
 ①01101011
 1

3 → 01111100
 00100100

3 → 10100100
 10000100

4 → 10000100
 ①00000001
 1
 00100101

3 → 10100000
 10000100

①00100000
 1
 00100001

1 → 10011001

2 → + 11100010
 ①01111011
 1

3 → 01111100
 00100100
 10100000

4 → + 10000100
 ①00100100
 1

00100101 ← Sum

10011001

+ 11100010
 101111011
 1

10111100
 + 00100100
 000000

Complimented data of sum

00100101 ✓
11011010 ⇨ checksum

Receiver side:-

1 → 10011001

2 → 11100010

10111101

= Same as
Sender.

00100101 → sum

11011010 → checksum

Add both

11111111 → sum

Complement sum

11111111

00000000

Accepted

error free

Q7

Msg :- 11001001

CRC poly :- $x^3 + 1 =$

$$1 \cdot x^3 + 0 \cdot x^2 + 0 \cdot x^1 + 1 \cdot x^0$$

CRC generator $\rightarrow$ 1001

Since poly is of order 3
append 3 0's at end of msg

Msg :- 1001 | 11001001000

1001	↓					
1011						
1001	↓					
1000						
1001	↓					
1100						
1001	↓					
1010						

1001

0011 $\rightarrow$ Rem $\rightarrow$ 011transmitted message $\Rightarrow$ 1101001011

DATE: _____
Why Subnetting is a need?

→ crucial in networking for optimizing IP address usage, enhancing network management.

Why essential ?????

- 1) Efficient IP Address Allocation
- 2) Enhanced Network Management
- 3) Improved Security

Find class of each IP address

1) 227.12.14.87

→ class D address used for multicast.

2) 193.14.56.22

→ first octet falls within range of 192-223, it's class C address.

3) 14.23.120.81

→ range 1-126.

→ class A address

4) 252.5.15.111

⇒ 240-255

⇒ class E address

5) 134.11.78.56

128-191 → class B address.